makes use of Trivers's 'cruel bind' idea, referred to on page 193, as follows.

Many fish do not copulate, but instead simply spew out their sex cells into the water. Fertilization takes place in the open water, not inside the body of one of the partners. This is probably how sexual reproduction first began. Land animals like birds, mammals and reptiles, on the other hand, cannot afford this kind of external fertilization, because their sex cells are too vulnerable to drying-up. The gametes of one sex—the male, since sperms are mobile—are introduced into the wet interior of a member of the other sex—the female. So much is just fact. Now comes the idea. After copulation, the land-dwelling female is left in physical possession of the embryo. It is inside her body. Even if she lays the fertilized egg almost immediately, the male still has time to vanish, thereby forcing the female into Trivers's 'cruel bind'. The male is inevitably provided with an opportunity to take the prior decision to desert, closing the female's options, and forcing her to decide whether to leave the young to certain death, or whether to stay with it and rear it. Therefore, maternal care is more common among land animals than paternal care.

But for fish and other water-dwelling animals things are very different. If the male does not physically introduce his sperms into the female's body there is no necessary sense in which the female is left 'holding the baby'. Either partner might make a quick getaway and leave the other one in possession of the newly fertilized eggs. But there is even a possible reason why it might often be the male who is most vulnerable to being deserted. It seems probable that an evolutionary battle will develop over who sheds their sex cells first. The partner who does so has the advantage that he or she can then leave the other one in possession of the new embryos. On the other hand, the partner who spawns first runs the risk that his prospective partner may subsequently fail to follow suit. Now the male is more vulnerable here, if only because sperms are lighter and more likely to diffuse than eggs. If a female spawns too early, i.e. before the male is ready, it will not greatly matter because the eggs, being relatively large and heavy, are likely to stay together as a coherent clutch for some time. Therefore a female fish can afford to take the 'risk' of spawning early. The male dare not take this risk, since if he spawns too early his sperms will have diffused away before the female is ready, and she will then not spawn herself, because it will not be worth her